Induction Proofs

To prove $\forall n \in \mathbb{N} : P(n)$, that is, that $P(n)$ is true for all natural numbers $n \geq 0$, it suffices to prove two things:

(a) Prove $P(0)$ (*base case*) *and*

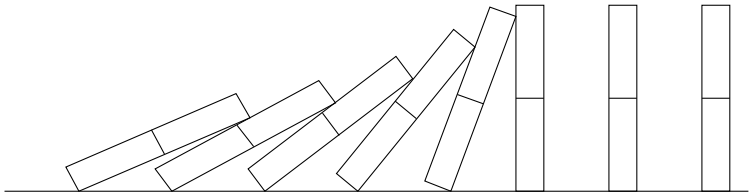
(b) Prove $\forall k \in \mathbb{N} \left(P(k) \Rightarrow P(k+1) \right)$ (*induction step*)

This is the principle of ***(mathematical) induction***.

For a given property $P(n)$ of numbers, if we can show the two things above, then $P(n)$ must be true of all $n \in \mathbb{N}$:

- By (a), $P(0)$ must be true.
- Since $P(0)$ is true, by (b), $P(1)$ must be true.
- Since $P(1)$ is true, by (b), $P(2)$ must be true.
- Since $P(2)$ is true, by (b), $P(3)$ must be true.

⋮ ⋮ ⋮



A First Proof by Induction

Theorem: $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ for all $n \geq 0$.

Proof: By induction on n.

- For the base case ($n=0$), we note that

  $$1 + 2 + \cdots + 0 = 0 = \frac{0(0+1)}{2}.$$

- For the induction step, we assume that for *some* k

  $$1 + 2 + \cdots + k = \frac{k(k+1)}{2}$$

  and prove that $1 + 2 + \cdots + k + (k+1) = \frac{(k+1)(k+2)}{2}$

  This is shown so:

  $$\underbrace{1 + 2 + \cdots + k} + (k+1)$$

  $$= \frac{k(k+1)}{2} + (k+1) \qquad \textit{(by the inductive hypothesis)}$$

  $$= \frac{(k+1)(k+2)}{2}$$

□

Induction via Contradiction

Theorem: $1 + 2 + \cdots + n = \frac{n(n+1)}{2}$ for all $n \geq 0$.

Proof: By *contradiction*.

Suppose that for *some* $n \geq 0$, $1 + 2 + \cdots + n \neq \frac{n(n+1)}{2}$.

Let n be the *least* number for which the theorem fails;
that is, suppose the theorem is true for all values less than n.

We argue by cases on the "structure" of n:

$n = 0$: $1 + 2 + \cdots + 0 = 0 = \frac{0(0+1)}{2} = \frac{n(n+1)}{2}$ *contradiction!*

$n = k+1$: $1 + 2 + \cdots + n$

$$= \underbrace{1 + 2 + \cdots + k} + (k+1)$$

$$= \frac{k(k+1)}{2} + (k+1) \quad \textit{(by assumption)}$$

$$= \frac{(k+1)(k+2)}{2} = \frac{n(n+1)}{2}$$

contradiction! □

Another Proof by Induction

Theorem: $1^2 + 2^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$ for all $n \geq 0$.

Proof: By induction on n.

- For the base case ($n=0$), we note that

$$1^2 + 2^2 + \cdots + 0^2 \;=\; 0 \;=\; \frac{0(0+1)(2\cdot 0+1)}{6}$$

- For the induction step, we assume that for *some* k

$$1^2 + 2^2 + \cdots + k^2 \;=\; \frac{k(k+1)(2k+1)}{6}$$

and prove that $1^2 + 2^2 + \cdots + k^2 + (k+1)^2 \;=\; \frac{(k+1)(k+2)(2k+3)}{6}$.

This is shown so:

$$\underbrace{1^2 + 2^2 + \cdots + k^2} + (k+1)^2$$

$$= \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \qquad \textit{(by the inductive hypothesis)}$$

$$= \frac{(k+1)(k+2)(2k+3)}{6}$$

$6^n + 4$ is divisible by 10

Theorem: $6^n + 4$ is divisible by 10, for every integer $n > 0$.

Proof: By induction on n.

- For the base case ($n=1$), $6^1+4 = 10$ is divisible by 10.
- For the induction step, we assume that $6^k + 4$ is divisible by 10 and from this show that $6^{k+1} + 4$ must also be divisible by 10.

  So suppose that $6^k + 4$ is divisible by 10, say

  $$6^k + 4 = 10m \quad \text{for some integer } m.$$

  Then
  $$\begin{aligned} 6^{k+1} + 4 &= 6(6^k) + 4 \\ &= 6(6^k+4) - 20 \\ &= 6(10m) - 20 \\ &= 10(6m - 2). \end{aligned}$$

  That is, $6^{k+1} + 4$ is divisible by 10 as required. □

A Fact About Fibonacci Numbers

Theorem: $f_1 + f_2 + \cdots + f_n = f_{n+2} - 1$ for all $n \geq 0$.

(Recall that $f_0 = 0$, $f_1 = 1$ and $f_n = f_{n-1} + f_{n-2}$.)

Proof: By induction on n.

- For the base case ($n=0$),

$$f_1 + f_2 + \cdots + f_0 = 0 = 1 - 1 = f_2 - 1.$$

- For the induction step, we assume that the statement of the Theorem is true for $n=k$, and from this show that it is true for $n=k+1$.

$$\begin{aligned} f_1 + f_2 + \cdots + f_{k+1} &= \left(f_1 + f_2 + \cdots + f_k\right) + \left(f_{k+1}\right) \\ &= \left(f_{k+2} - 1\right) + \left(f_{k+1}\right) \qquad \textit{(by induction)} \\ &= \left(f_{k+2} + f_{k+1}\right) - 1 = f_{k+3} - 1 \end{aligned}$$

□

A Fact About Harmonic Numbers

Theorem: $H_1 + H_2 + \cdots + H_n = (n+1)H_n - n$ for all $n \geq 0$.

(Recall that $H_0 = 0$ and $H_n = H_{n-1} + \frac{1}{n}$.) **Proof**: By induction on n.

- For the base case ($n=0$),

$$H_1 + H_2 + \cdots + H_0 = 0 = (0+1)H_0 - 0.$$

- For the induction step, we assume that the statement of the Theorem is true for $n=k$, and from this show that it is true for $n=k+1$.

$$\underbrace{H_1 + H_2 + \cdots + H_k} + H_{k+1}$$

$$= \overbrace{\left((k+1)\underbrace{H_k} - k\right)} + H_{k+1} \qquad \text{(by induction)}$$

$$= \left(\underbrace{(k+1)\overbrace{\left(H_{k+1} - \tfrac{1}{k+1}\right)}} - k\right) + H_{k+1}$$

$$= \left(\overbrace{(k+1)H_{k+1} - 1} - k\right) + H_{k+1} = (k+2)H_{k+1} - (k+1) \qquad \square$$

Sizes of Power Sets

Theorem: If A is a finite set with n elements,

then its powerset $\mathcal{P}(A)$ has 2^n elements.

Proof: By induction on n.

- For the base case, if A has 0 elements, then $A = \emptyset$, and $\mathcal{P}(\emptyset) = \{\emptyset\}$ has $1 = 2^0$ elements.
- For the induction step, we assume that the power set of any set with k elements has 2^k elements, and we prove that the power set of any set with $k+1$ elements must have 2^{k+1} elements.

  Let A be any set with $k+1$ elements, and let $x \in A$; then $B = A \setminus \{x\}$ has k elements, and so by the inductive hypothesis, $\mathcal{P}(B)$ has 2^k elements.

  A subset of A is either a subset of B (of which there are 2^k), or a subset of B with the element x added (of which there are again 2^k).

  Thus we get $2^k + 2^k = 2^{k+1}$ elements of $\mathcal{P}(A)$. $\square$

Strong Induction

To prove $\forall n \in \mathbb{N} : P(n)$, that is, that $P(n)$ is true for all natural numbers $n \geq 0$, it suffices to prove that $P(n)$ holds under the assumption that $P(k)$ holds for every $k<n$.

Example: Let $f(n) = \begin{cases} 0, & \text{if } n=0; \\ 2 \cdot f(n/2), & \text{if } n>0 \text{ even}; \\ f(n-1)+1, & \text{if } n \text{ odd}. \end{cases}$

Theorem: $f(n) = n$ for every $n \geq 0$.

Proof: By (strong) induction on n, arguing by cases on the "structure" of n.

$\underline{n=0}$**:** $f(0) = 0$.

$\underline{n>0 \textbf{ even}}$**:** $f(n) = 2 \cdot f(n/2) = 2 \cdot (n/2) = n$

$\underline{n \textbf{ odd}}$**:** $f(n) = f(n-1)+1 = (n-1)+1 = n$

Fundamental Theorem of Arithmetic

Theorem: Every integer $n > 1$ is either prime or a product of primes.

Proof: By (strong) induction on n.

Suppose that $n>1$, and that for every integer k with $1<k<n$, k is either prime or the product of primes.

If n itself is prime then we have nothing to prove.

Thus, suppose that $n = ab$ with $1<a<n$ and $1<b<n$.

By the inductive hypothesis, each of a and b is either prime or a product of primes.

But then since $n = ab$, n itself is a product of primes. $\square$

Induction over Words

Recall the inductive definition of a word over $\{a, b\}$: $w ::= \varepsilon \mid aw \mid bw$

That is, a word is either the empty word ε, or it is a nonempty word aw or bw consisting of either an a or a b followed by a (smaller) word w.

The ***length*** of a word is inductively defined by:

$$length(\varepsilon) = 0 \qquad length(as) = 1 + length(s) \qquad length(bs) = 1 + length(s)$$

Theorem: $aw \neq wb$ for all words w.

Proof: By induction on the structure of w. (That is, by induction on the length of w, arguing by cases on the structure of w.)

$\underline{w = \varepsilon}$**:** $aw = a \neq b = wb$.

$\underline{w = as}$**:** By induction, $as \neq sb$, so $aas \neq asb$.

But then $aw = aas \neq asb = wb$.

$\underline{w = bs}$**:** $aw = abs \neq bsb = wb$. □

Induction over Trees

Recall the inductive definition of a binary tree: $t ::= \star \mid N(t_1, t_2)$

That is, a binary tree is either a leaf $\star$, or an internal node $N(t_1, t_2)$ with two (smaller) trees t_1 and t_2.

Theorem: Every tree t has exactly one more leaf than (internal) node.

Proof: By induction on the structure of t. (That is, by induction on the size of t, arguing by cases on the structure of t.)

$\underline{t = \star}$**:** $\star$ has 1 leaf and 0 nodes.

$\underline{t = N(t_1, t_2)}$**:** By induction, t_i $(i = 1, 2)$ must have n_i nodes and $n_i + 1$ leaves, for some n_1, n_2.

But then $N(t_1, t_2)$ must have n_1+n_2+1 nodes

and $(n_1+1) + (n_2+1) = (n_1+n_2+1)+1$ leaves. □

A Proof About Postage Stamps

Theorem: Any amount of postage of at least 8 pence can be made up from just 3-pence and 5-pence stamps.

Proof: The base case is straightforward: a 3-pence stamp and a 5-pence stamp make up 8 pence.

For the induction step, assume that we have a collection of such stamps adding up to a total of $n \geq 8$ pence.

- if there is a 5-pence stamp in this collection, remove it and replace it with two 3-pence stamps;
- If there are no 5-pence stamps, then there must be (at least) three 3-pence stamps in the collection; remove these and replace them with two 5-pence stamps.

In either case, we arrive at a collection of stamps adding up to $(n+1)$ pence.

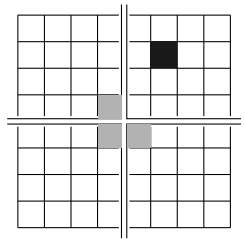
A Proof About Playing Boards

You are given a $2^n \times 2^n$ checkerboard with one black square arbitrarily placed on the board and the remaining $4^n - 1$ squares white. You are also given a supply of tiles which look like 2×2 checkerboards with one corner square removed. You want to tile the checkerboard so that each white square is covered exactly once, while the black square remains uncovered.

Theorem: The checkerboard can be so tiled.

Proof: By induction on n. For the base case, the 1×1 checkerboard can be tiled with 0 tiles.

For the induction step, place the first tile in the center of the board with the gap in the quadrant containing the black square.



By the inductive hypothesis, the four $2^{n-1} \times 2^{n-1}$ sub-quadrants can be tiled. □

A Non-Proof About Sums

"Non-Theorem": $1 + 2 + 3 + \cdots + n = \frac{(n-1)(n+2)}{2}$.

"Non-Proof": By induction on n.

$$\underbrace{1 + 2 + 3 + \cdots + k} + (k+1)$$

$$= \frac{(k-1)(k+2)}{2} + (k+1)$$

(by the inductive hypothesis)

$$= \frac{k(k+3)}{2}.$$ $\square$

Question: What is wrong with this argument?

A Non-Proof About People

"Non-Theorem": All people are the same age.

"Non-Proof": We prove, by induction on n, that for every set S of $n \geq 0$ people, all people in S are the same age.

For the base case, let $S{=}\emptyset$ consist of 0 people; trivially the claim holds in this instance.

For the induction step, assume that the claim holds for all sets of size less than n.

Let S be a set of n people. Let S' and S'' be two overlapping sets of people which together make up S: $S = S' \cup S''$ and $S' \cap S'' \neq \emptyset$.

By the inductive hypothesis, all people in S' are the same age, and all people in S'' are the same age. As S' and S'' overlap, all people in S must be the same age. □

Question: What is wrong with this argument?